```python
# HR style heading

print("="*70)
print("        ELECTRIC VEHICLE SALES ANALYSIS - INDIA")
print("             CODEALPHA DATA VISUALIZATION")
print("="*70)
```

```
======================================================================
        ELECTRIC VEHICLE SALES ANALYSIS - INDIA
             CODEALPHA DATA VISUALIZATION
======================================================================
```

```python
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")

df = pd.read_csv("Electric Vehicle Sales by State in India.csv")

df.head()
```


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```
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```

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```
df.shape
```

```
(96845, 8)
```

```
df.columns
```

```
Index(['Year', 'Month_Name', 'Date', 'State', 'Vehicle_Class',
       'Vehicle_Category', 'Vehicle_Type', 'EV_Sales_Quantity'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 96845 entries, 0 to 96844
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Year               96845 non-null  float64
 1   Month_Name         96845 non-null  object
 2   Date               96845 non-null  datetime64[ns]
 3   State              96845 non-null  object
 4   Vehicle_Class      96845 non-null  object
 5   Vehicle_Category   96845 non-null  object
 6   Vehicle_Type       96845 non-null  object
 7   EV_Sales_Quantity  96845 non-null  float64
dtypes: datetime64[ns](1), float64(2), object(5)
memory usage: 5.9+ MB
```

```
df.describe()
```

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```
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],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    }\n  ]\n}","type":"dataframe"}
```


```
df.isnull().sum()
```

```
Year                 0
Month_Name           0
Date                 0
State                0
Vehicle_Class        0
Vehicle_Category     0
Vehicle_Type         0
EV_Sales_Quantity    0
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df["Date"] = pd.to_datetime(df["Date"])
```

```
df.head()
```


```
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```

```
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}\n    },\n    {\n      \"column\": \"Vehicle_Class\",\n
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\"RECOVERY VEHICLE\"\n        ],\n        \"semantic_type\": \"\",\n
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Wheelers\"\n        ],\n        \"semantic_type\": \"\",\n
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\"2W_Shared\",\n          \"Others\"\n        ],\n
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n    }\n  ]\n}","type":"dataframe","variable_name":"df"}
```


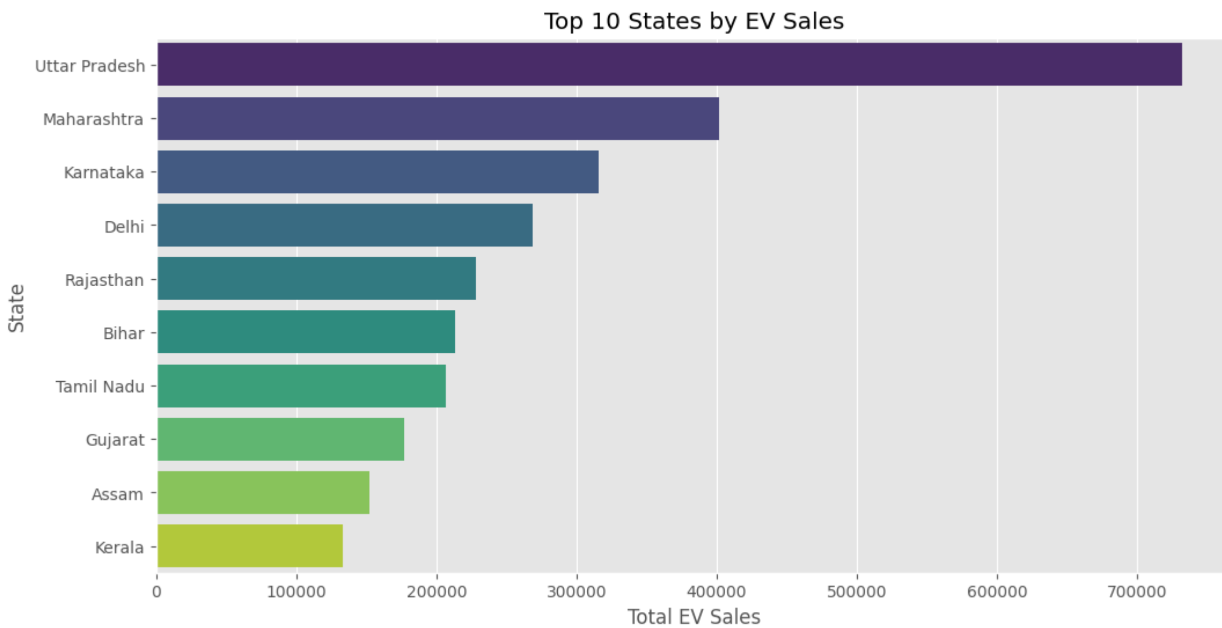
Data Visualization

Which states have the highest EV Sales?

```
state_sales = (
    df.groupby("State")["EV_Sales_Quantity"]
    .sum()
    .sort_values(ascending=False)
)

plt.figure(figsize=(12,6))

sns.barplot(
    x=state_sales.head(10).values,
    y=state_sales.head(10).index,
    palette="viridis"
)
```

```
plt.title("Top 10 States by EV Sales")

plt.xlabel("Total EV Sales")

plt.ylabel("State")

plt.show()
```

```
/tmp/ipykernel_738/3777386884.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `y` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(
```

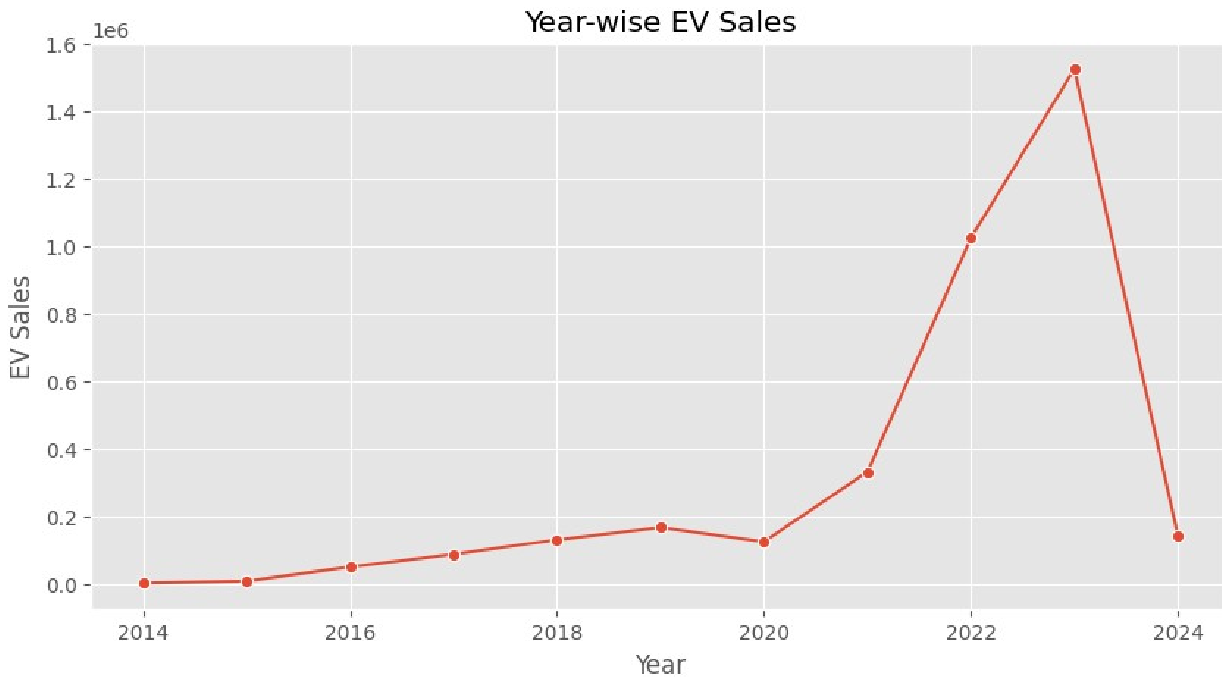


Uttarpradesh has the highest EV sales amongst all other states

How have EV Sales changed over the years?

```
year_sales = (
    df.groupby("Year")["EV_Sales_Quantity"]
    .sum()
)

plt.figure(figsize=(10,5))
```

```
sns.lineplot(
    x=year_sales.index,
    y=year_sales.values,
    marker="o"
)

plt.title("Year-wise EV Sales")

plt.xlabel("Year")

plt.ylabel("EV Sales")

plt.show()
```



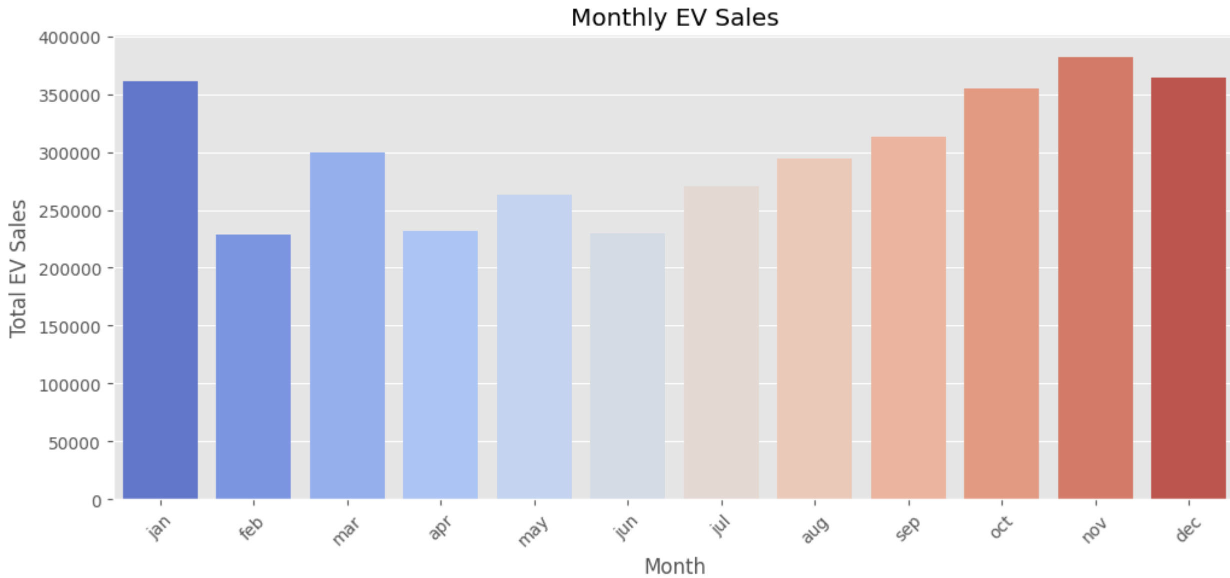
The data shows stable growth during 2014–2020, followed by a period of rapid expansion from 2020–2023. After reaching its highest level in 2023, the trend declined sharply in 2024, marking a significant year-over-year decrease

Which month records the highest EV Sales?

```
# Business Question 3
# Which month records the highest EV Sales?

month_sales = (
    df.groupby("Month_Name")["EV_Sales_Quantity"]
      .sum()
)

# Correct order based on your dataset
month_order = [
    "jan","feb","mar","apr","may","jun",
    "jul","aug","sep","oct","nov","dec"
]

month_sales = month_sales.reindex(month_order)

plt.figure(figsize=(12,5))

sns.barplot(
    x=month_sales.index,
    y=month_sales.values,
    palette="coolwarm"
)

plt.title("Monthly EV Sales")
plt.xlabel("Month")
plt.ylabel("Total EV Sales")

plt.xticks(rotation=45)

plt.show()
```

```
/tmp/ipykernel_738/3833595884.py:19: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.
```

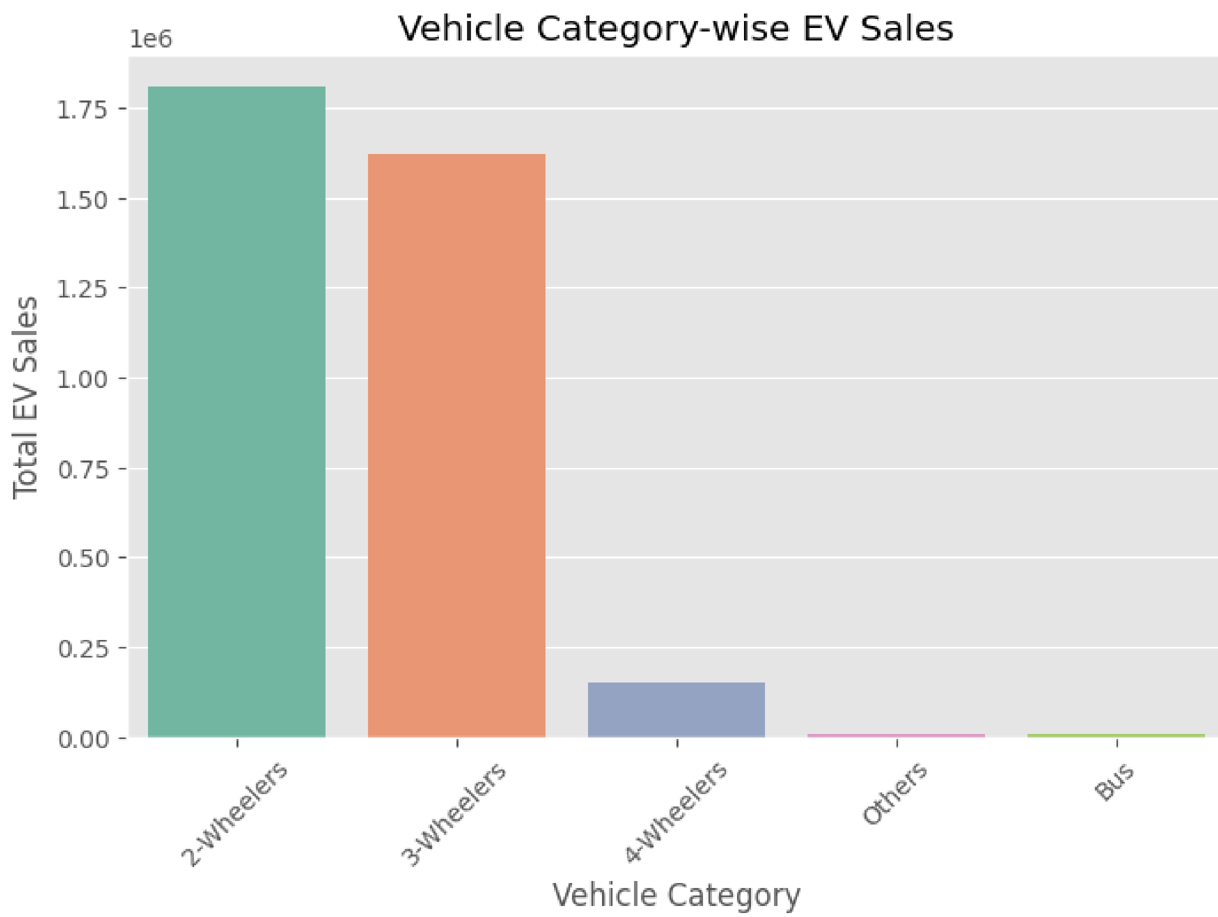
```
sns.barplot(
```



The analysis shows that EV sales peaked in November, while December and January recorded the second and third highest sales, respectively. This suggests that consumer demand for electric vehicles was strongest during the November period.

```
category_sales = (
    df.groupby("Vehicle_Category")["EV_Sales_Quantity"]
      .sum()
      .sort_values(ascending=False)
)

plt.figure(figsize=(8,5))

sns.barplot(
    x=category_sales.index,
    y=category_sales.values,
    palette="Set2"
)

plt.title("Vehicle Category-wise EV Sales")
plt.xlabel("Vehicle Category")
plt.ylabel("Total EV Sales")

plt.xticks(rotation=45)
```

```
plt.show()

/tmp/ipykernel_738/1435507319.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(
```



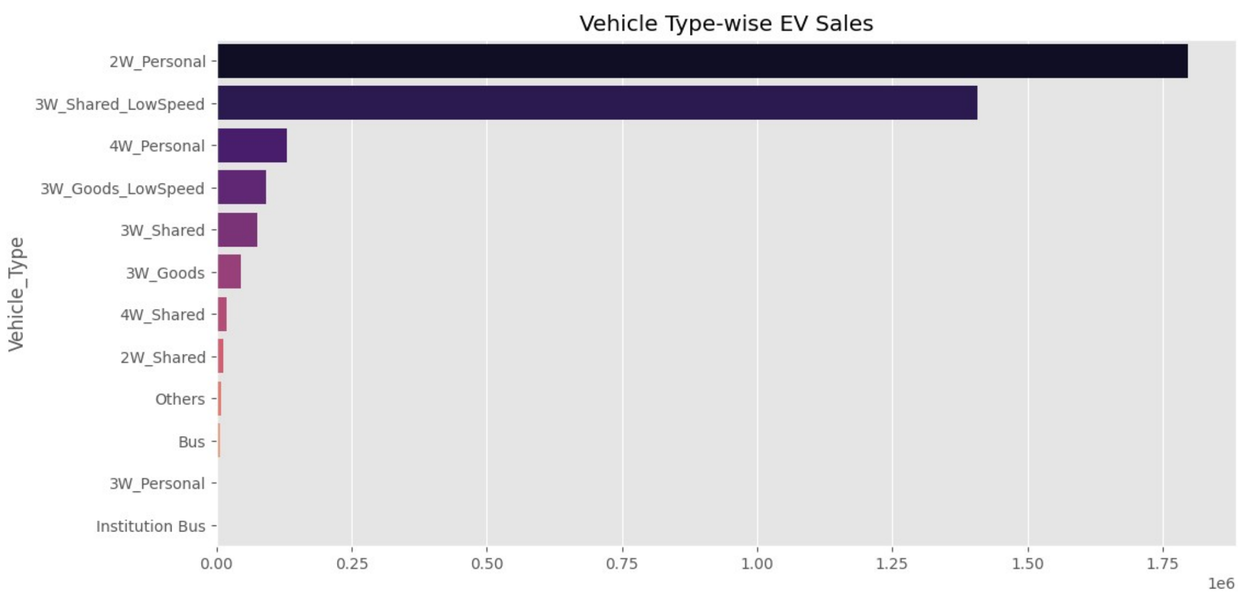
The 2-wheelers category has the highest EV Sales amongst all vehicle categories

Which Vehicle Type has the highest EV Sales?

```
vehicle_type = (
    df.groupby("Vehicle_Type")["EV_Sales_Quantity"]
    .sum()
```

```
    .sort_values(ascending=False)
)

plt.figure(figsize=(12,6))

sns.barplot(
    x=vehicle_type.values,
    y=vehicle_type.index,
    palette="magma"
)

plt.title("Vehicle Type-wise EV Sales")

plt.show()
```

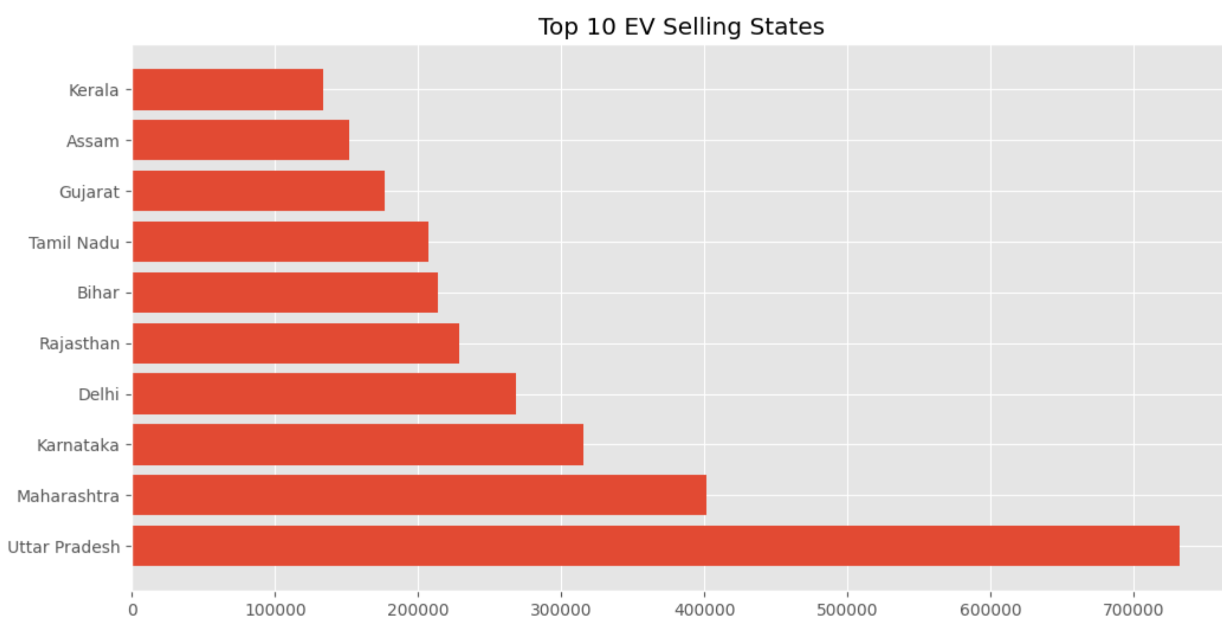
```
/tmp/ipykernel_738/2570376533.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `y` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(
```



Amongst all vehicle types the 2W-Personal has the highest EV sales

Top 10 States by EV Sales

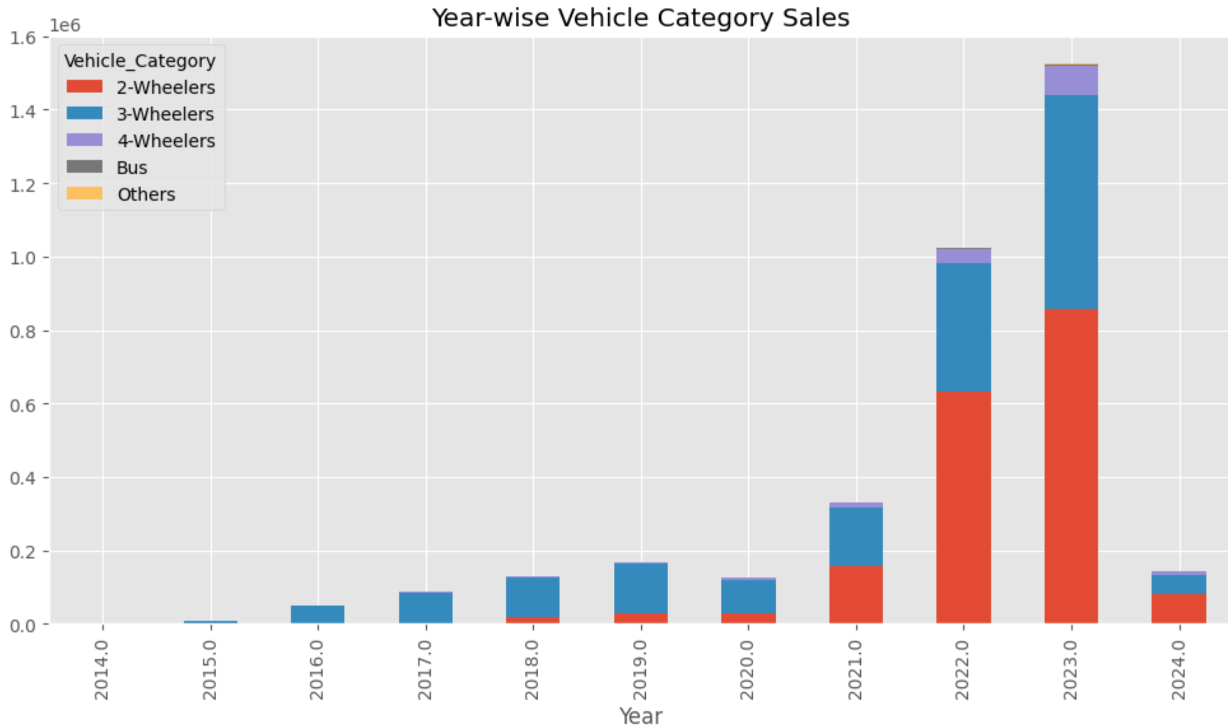
```
top10 = state_sales.head(10)

plt.figure(figsize=(12,6))

plt.barh(top10.index,top10.values)

plt.title("Top 10 EV Selling States")

plt.show()
```



Year-wise Vehicle Category Sales

```
pivot = df.pivot_table(
    values="EV_Sales_Quantity",
    index="Year",
    columns="Vehicle_Category",
    aggfunc="sum"
)

pivot.plot(
    figsize=(12,6),
    kind="bar",
```

```
    stacked=True
)

plt.title("Year-wise Vehicle Category Sales")

plt.show()
```

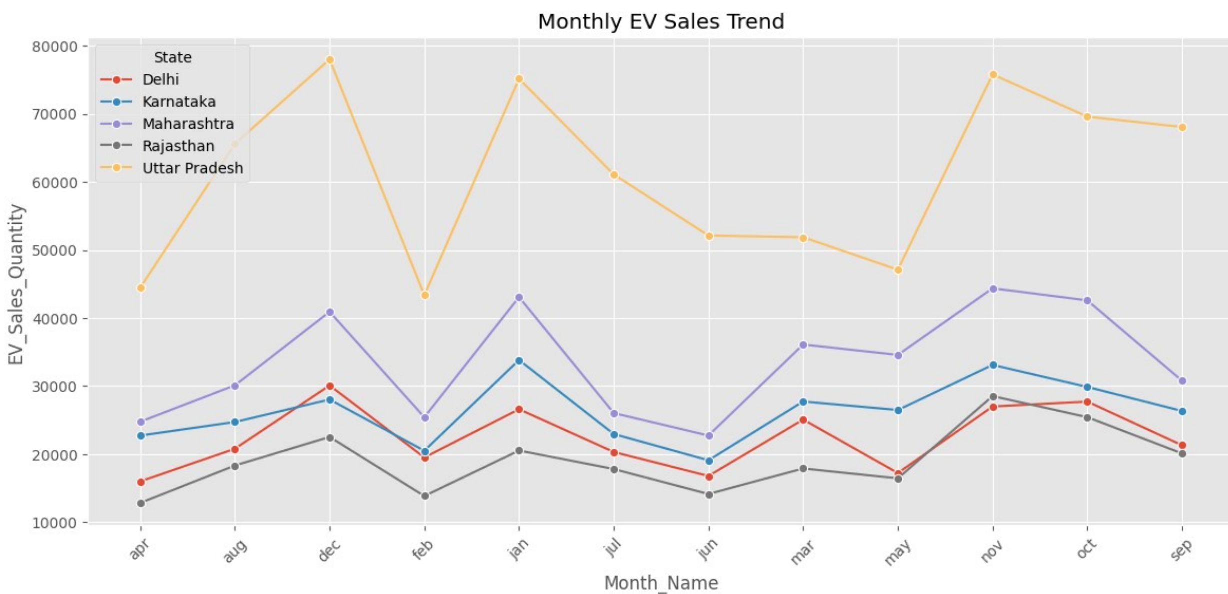


The analysis reveals that 2023 recorded the highest Electric Vehicle (EV) sales, followed by 2022. This indicates a significant increase in EV adoption during 2023, reflecting growing consumer awareness, supportive government policies, and improved EV infrastructure.

State-wise Monthly EV Sales Trend

```
state_month = df.groupby(
    ["Month_Name","State"]
)["EV_Sales_Quantity"].sum().reset_index()

top_states = state_sales.head(5).index

filtered = state_month[
```

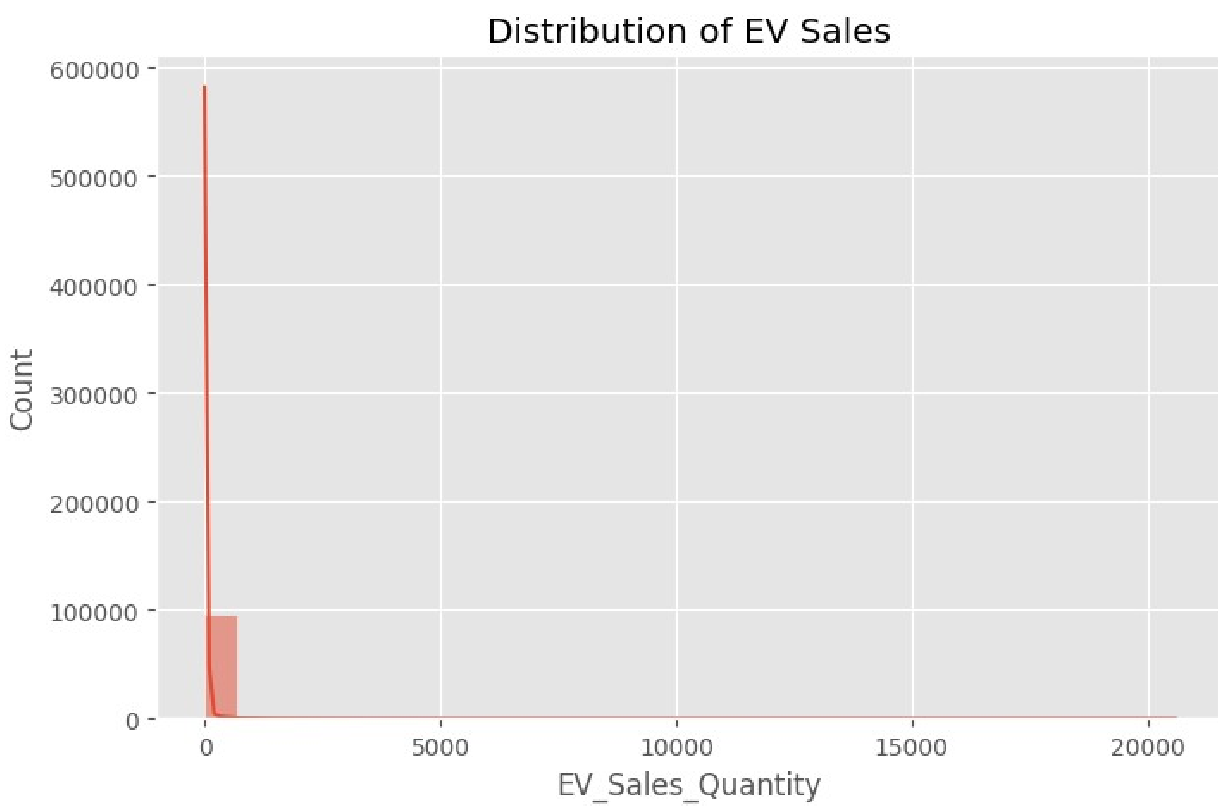
```
    state_month["State"].isin(top_states)
]

plt.figure(figsize=(14,6))

sns.lineplot(
    data=filtered,
    x="Month_Name",
    y="EV_Sales_Quantity",
    hue="State",
    marker="o"
)

plt.xticks(rotation=45)

plt.title("Monthly EV Sales Trend")

plt.show()
```



Uttar Pradesh recorded the highest Electric Vehicle (EV) sales among all states. This indicates a strong adoption of electric vehicles in the state, likely driven by its large population, increasing consumer demand, and supportive government initiatives.

Distribution of EV Sales

```
plt.figure(figsize=(8,5))

sns.histplot(
    df["EV_Sales_Quantity"],
    bins=30,
    kde=True
)

plt.title("Distribution of EV Sales")

plt.show()
```



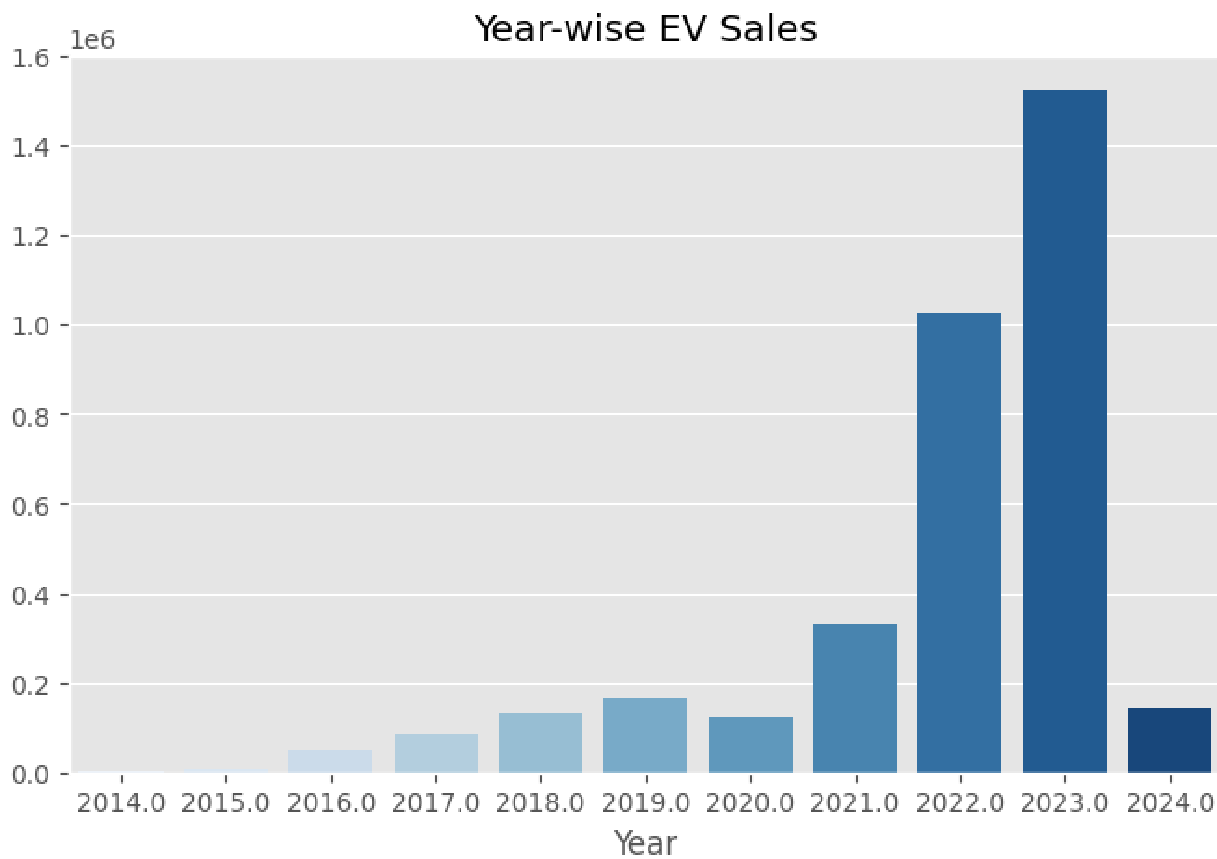
The distribution of EV sales is highly right-skewed, with the majority of records concentrated at lower EV sales quantities. Only a small number of observations have very high sales values, creating a long tail toward the right side of the distribution.

Looking at your histogram, the distribution is highly positively skewed (right-skewed).

Which Year recorded the highest EV Sales?

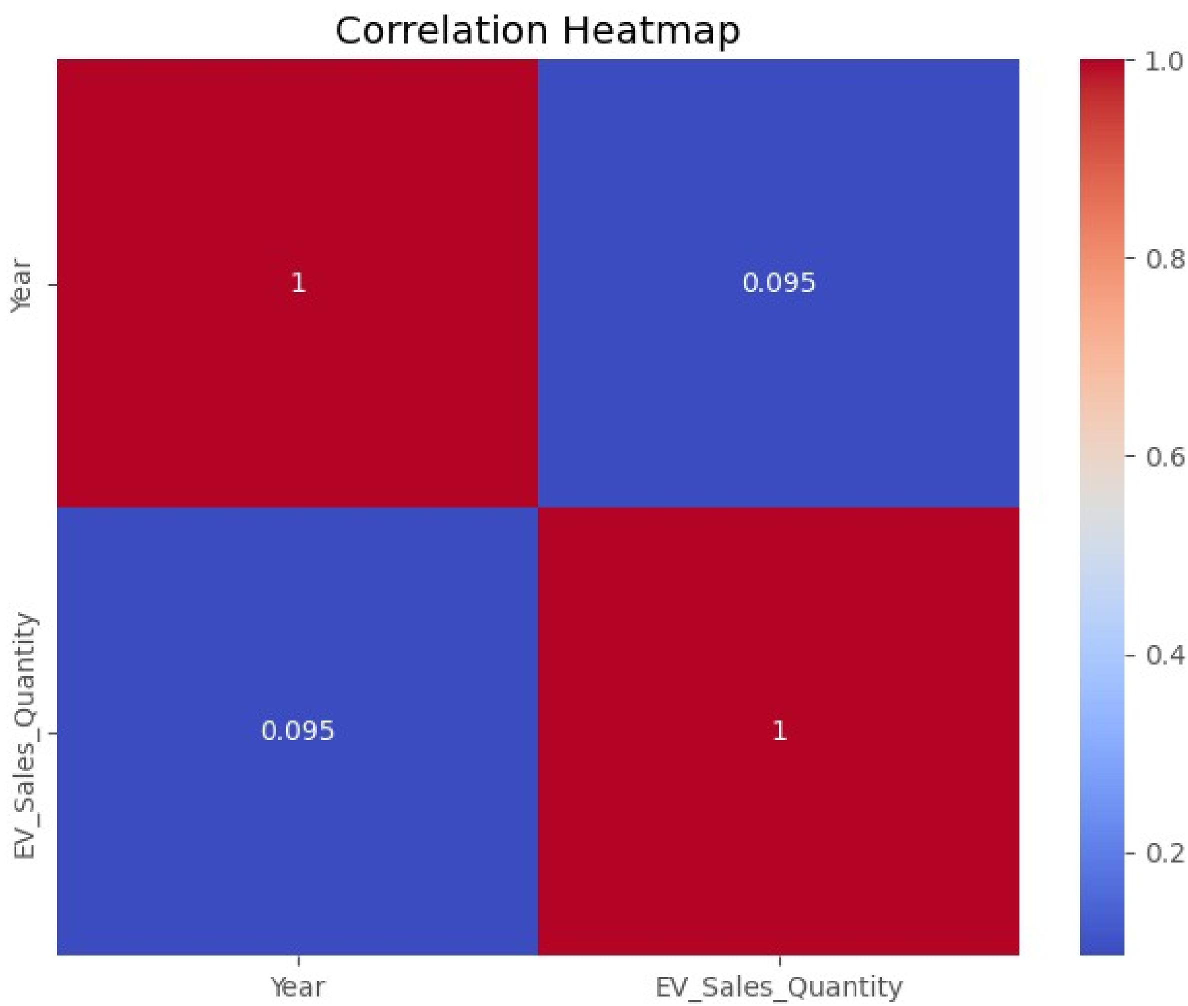
```
plt.figure(figsize=(8,5))

sns.barplot(
    x=year_sales.index,
    y=year_sales.values,
    palette="Blues"
)

plt.title("Year-wise EV Sales")

plt.show()
```

```
/tmp/ipykernel_738/1493620711.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(
```



Correlation Heatmap

```
plt.figure(figsize=(8,6))

numeric = df.select_dtypes(include=np.number)

sns.heatmap(
    numeric.corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")

plt.show()
```


Correlation Heatmap
Year
EV_Sales_Quantity
1
0.095
0.095
1
Year
EV_Sales_Quantity
1.0
0.8
0.6
0.4
0.2

Based on your correlation heatmap,the correlation heatmap indicates a very weak positive correlation (0.095) between Year and EV Sales Quantity. This suggests that EV sales have shown only a slight increase over the years, and the year alone does not strongly influence the sales quantity.

Overall, the heatmap reveals that there is no strong linear relationship between Year and EV Sales Quantity. Therefore, EV sales are influenced by multiple external factors, highlighting the need for a more comprehensive analysis beyond time-based trends.

Business Recommendations

Strengthen EV Infrastructure in Low-Sales States

Expand charging stations and service centers in states with lower EV adoption to encourage wider usage.

Leverage High-Performing Markets

###States such as Uttar Pradesh can be used as benchmark markets to identify successful policies and strategies that can be implemented in other regions.

Increase Promotional Campaigns During Peak Months

###Since November, December, and January recorded the highest EV sales, manufacturers and dealers should introduce festive offers, financing options, and marketing campaigns during these months to maximize sales.

Focus on High-Growth Years

###The strong sales growth observed in 2023 indicates increasing consumer acceptance of EVs. Companies should capitalize on this momentum by expanding production and launching new EV models.

Promote High-Demand Vehicle Categories

Manufacturers should invest more in the vehicle categories and types with the highest sales while improving affordability and availability in lower-performing segments.

Improve Consumer Awareness

Conduct awareness programs highlighting the environmental and economic benefits of electric vehicles to increase adoption in regions with lower sales.

Strengthen Government Incentives

Continue or enhance subsidies, tax benefits, and financing schemes to encourage both individual and commercial EV purchases.

Expand Charging Infrastructure

Increase the number of fast-charging stations along highways, urban centers, and rural areas to reduce range anxiety and improve customer confidence.

Use Data-Driven Decision Making

Regularly analyze EV sales trends by state, month, vehicle category, and vehicle type to optimize production, inventory management, and marketing strategies.

Collaborate with Public and Private Stakeholders

Encourage partnerships between government agencies, automobile manufacturers, charging network providers, and financial institutions to accelerate EV adoption across India.

Overall Conclusion

The data visualization analysis indicates that India's Electric Vehicle market is experiencing steady growth, with 2023 recording the highest sales and Uttar Pradesh emerging as the leading state in EV sales. Sales were highest during November, followed by December and January, suggesting strong seasonal demand. While EV adoption continues to increase, the distribution of sales remains uneven across states and vehicle segments. Expanding charging infrastructure, strengthening government incentives, promoting consumer awareness, and focusing on data-driven strategies will be essential to achieving balanced and sustainable EV adoption across the country